

Cadmus return on investment estimation

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This artwork was developed by Marcus Lee Design to reflect Nous Group's Reconciliation Action Plan and our aspirations for respectful and productive engagement with Aboriginal and Torres Strait Islander peoples and communities.

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Executive summary

Cadmus partners with universities globally to deliver secure, authentic, and scaffolded academic assessments, with pilot studies indicating positive outcomes for students and universities. Studies have observed associations between Cadmus usage and student engagement, academic performance, and reduced subject withdrawals for Cadmus-using cohorts. These observed improvements are expected to accumulate across the student journey and translate into meaningful benefits for students and an estimated average return on investment (ROI) of 7.1 for pilot universities.

Nous was commissioned by Cadmus to calculate the ROI of Cadmus services to universities, in relation to student retention and academic performance, and based on the outcomes observed in university pilots.

Cadmus reduces pre-census course withdrawals

Early subject withdrawals create immediate revenue loss for universities. Cadmus helps prevent withdrawals by improving assessment clarity, confidence, and early engagement.

Pilot studies show a consistent reduction in pre-census withdrawals across institutions due to Cadmus, with a median reduction of 2.7%. When combined with typical tuition fees, this reduction creates a significant financial benefit for universities each year.

Cadmus improves academic performance and likely longer-term student retention

Academic literature evidences that higher grades in early years strongly influence whether students stay enrolled and complete their degree. Cadmus improves grades (by 2.7% on average), which in turn increases the likelihood that students progress through their programme.

This analysis estimates the immediate and future value created for universities when more undergraduate students remain enrolled due to Cadmus, while reflecting a diminishing effect to students in latter years of study and the average cost of a retained student.

Improved student retention is attributed conservatively to Cadmus to reflect that multiple institutional factors and student characteristics contribute to retention.

Across pilots in Australia and the UK, universities are estimated to have realised a weighted average ROI of 7.1

The modelled ROI to a university using Cadmus comprises the estimated net annual financial benefit to a university compared to the annual subscription fee of Cadmus.

The average estimated ROI for a pilot Australian university is 8.6 and 4.3 for a UK university. Retention benefits comprise approximately 60% of total estimated benefits to each university. The ROI rises as Cadmus is deployed across more modules. Given that many universities have deployed more modules post-pilot, the ROI at those universities is expected to have already increased. ROI estimates exclude other non-quantified benefits, such as time savings and quality improvements for academics designing, marking, and managing assessments.

What this means for universities

Cadmus provides an assessment approach that is associated, in pilot settings, with improvements in student experience and academic outcomes, alongside likely financial benefits. The findings show that:

- Student success improves in meaningful, quantifiable ways
- Early withdrawals reduce, preserving tuition revenue
- Higher grades support stronger progression and completion
- Value grows significantly as more modules use Cadmus

These outcomes show that Cadmus is not only a learning tool—it is a strategic investment that supports student achievement and improves institutional sustainability.

Cadmus can further strengthen the robustness of future ROI estimates by expanding student-level retention data collection and incorporating outcomes from a broader range of universities, enabling more definitive causal analysis of its impact

Introduction

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University partners are identifying benefits delivered by Cadmus services.

Cadmus is an Australian-based assessment platform partnering with universities globally to deliver secure, authentic, and scaffolded assessments that improve student success and retention. Pilots at university partners have measured the early outcomes of Cadmus adoption and established a path to quantifying the value to universities.

Overview of Cadmus services

Cadmus provides an end-to-end assessment platform that supports students throughout the assessment process. It enables:

- Authentic assessment design with integrated resources and checklists.
- Scaffolded learning to build academic skills and confidence.
- Secure assessment delivery for exams and assignments, mitigating misconduct risks.

Cadmus "modules" refer to university courses, units or subjects in which Cadmus is used.

Studies on the impact and outcomes of Cadmus

Cadmus and its university partners have collected data from pilots and enterprise implementations between 2023 and 2025, comparing outcomes for students who used Cadmus versus those who worked without. Using this partner-supplied data and Cadmus platform analytics, Cadmus has derived results across multiple institutions to identify the positive impact on student success and retention.

The studies observed associations for students using Cadmus, including:

- Better academic performance (average grade improvements)
- Reduced course withdrawals
- Increase in submission rates
- Increase in student pass rates
- Improved student experience.

Observed outcomes of the implementation of Cadmus in pilots which are used in this ROI analysis are presented in the table to the right.

Table: Observable outcomes from the implementation of Cadmus in pilots at Australian and UK universities.

	Number of Cadmus modules[2]	Number of unique students using Cadmus	Reduction in course withdrawal rate[1]	Average grade improvement[1]
Australia University A, Globally Ranked, IRU University (~20k students)	5	467	2.5%	5.6%
Australia University B, Top-15 AUS Ranked University (~37k students)[3]	108	7,854	2.9%	2.7%
Australia University C, Leading Regional University (~18k students)	15	4,631	2.9%	1.6%
Australia University D, Nationally-Ranked Regional University (~18k students)	9	3,296	2.1%	2.5%
UK University A, Top-20 UK Ranked Russell Group University (~31k students)	63	4,200	*Not measured*	7.0%
UK University B, Top-20 UK Ranked University (~10k students)	10	317	*Not measured*	2.9%
UK University C, Top-5 UK Ranked Russell Group University (~51k students)	14	1,586	*Not measured*	1.4%

[1] Outcomes reflect correlations observed in pilots and not controlled causal analyses.
[2] Many universities have expanded the number of modules deployed since their pilot studies.
[3] Modules deployed and measured outcomes at Australia University B are post-pilot, as results were not measured during the pilot.

Cadmus commissioned Nous to calculate a quantifiable ROI.

With an opportunity to quantify the value of Cadmus to universities, Cadmus commissioned Nous to:

- Estimate the value of Cadmus to universities in relation to student retention and academic performance, and recommend how Cadmus can calculate a broader ROI for its services to university partners.

The approach included:

1. Conducting preliminary discussions with Cadmus to understand its services and university partnerships, and examining pilot evaluation reports to collect outcome data
2. Developing and testing model logic for ROI estimation (detailed on the following pages)
3. Collecting publicly available data on tuition fees, student numbers, and teaching costs to calculate the ROI
4. Documenting the methodology and results in this report.

ROI estimation rationale

Nous calculated the net impact of Cadmus on university finances based on available pilot data. We used pilot data on Cadmus' effect on course withdrawal rates and student grades to estimate the number of pre-Census course withdrawals avoided and the number of students retained in their degree (compared to a counterfactual of students not using Cadmus). We modelled estimated benefits as the savings from avoided course withdrawals and the lifetime value of retained students (less the marginal cost), with the net impact subtracting the costs of the Cadmus licence, outlined below. Attribution factors were applied to reflect uncertainty and avoid overstating Cadmus' contribution

QUANTIFIED FINANCIAL BENEFITS

- **Course withdrawal savings**: the model estimates the number of pre-census withdrawals prevented among Cadmus module enrolments, then values each prevented withdrawal using the weighted tuition fee per module.
- **Lifetime value of retained students**: the model estimates the uplift in degree-level retention attributable to Cadmus (derived from observed grade improvements), then values retained students using their future tuition revenue over remaining years of study (by year level).

QUANTIFIED FINANCIAL COSTS

- **Marginal cost of teaching and support for additional retained students**: the model subtracts an estimated marginal future teaching and administration cost associated with keeping each additional student enrolled, to convert additional revenue into a net surplus per retained student.
- **Cadmus licence cost:** the ROI incorporates the annual Cadmus subscription cost, which is subtracted from total benefits to calculate net benefit and ROI.

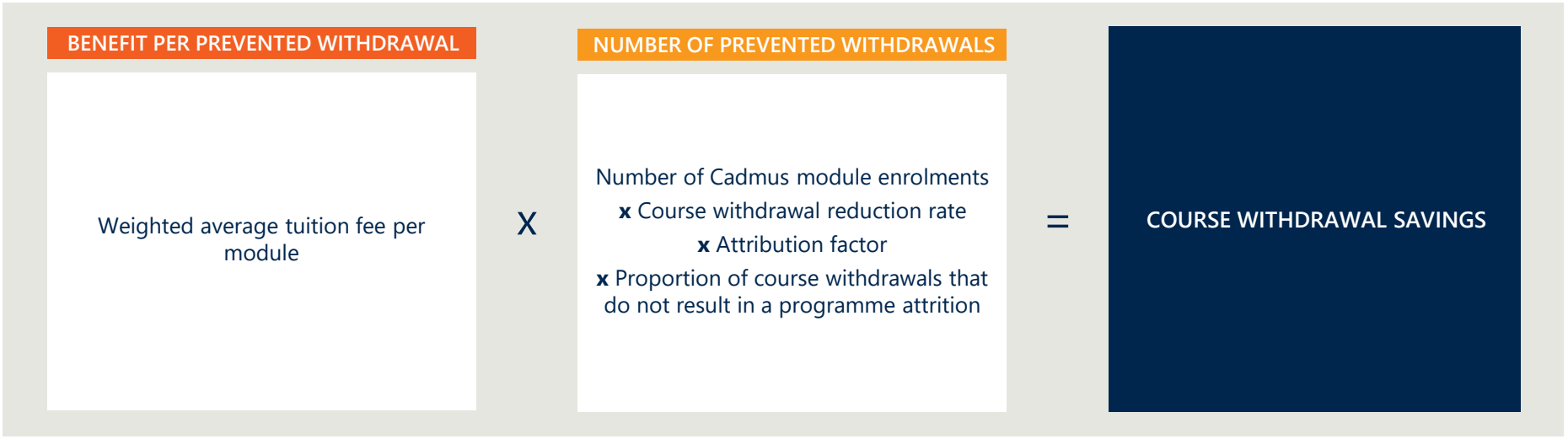
Benefit 1: Course withdrawal savings

We developed a calculation logic to quantify course withdrawal savings.

Course (subject or unit) withdrawals before the census date* create a direct financial impact for universities. Students can enrol and then withdraw up to the census date without incurring fees, which means tuition revenue is lost while many delivery costs remain fixed. These withdrawals often occur when students disengage early in the semester—commonly triggered by confusion about assessment requirements, lack of confidence, or feeling overwhelmed.

Cadmus is designed to help address these drivers by improving assessment clarity and structure, providing scaffolded tasks that break complex assignments into manageable steps, and enabling early engagement through integrated resources and progress nudges. This combination reduces uncertainty and supports students to start assessments sooner, building momentum and confidence. By strengthening engagement at critical early points in the semester, Cadmus helps prevent the late disengagement that leads to "drop before census" behaviour.

The logic model for calculating the course withdrawal savings is presented below.



The calculation details, assumptions, limitations and data sources for each of the components of the modelling logic are explained on the following page.

*A census date at a university is the official deadline in an academic term by which a student's course or subject enrolment is finalised. It's a key administrative cut-off with important academic and financial consequences. It is the last day on which a student can withdraw from a subject or course without academic penalty and without being financially liable for tuition fees.

Approach to calculating course withdrawal savings, assumptions and limitations.

	Benefit per prevented withdrawal	Number of withdrawals prevented
Calculation detail	**Weighted average tuition fee per module** • Australia: AUD $3,566; UK: £1,577. • Reflects blended domestic and international student programme-level fees, weighted by institutional student mix across domestic and international. • Domestic fees include government and student contributions (Australia) and regulated maximum fees (UK). • Divided into a per-module figure (total annual tuition ÷ 8 modules per student per year).	**Number of Cadmus module enrolments** • The number of students who have used Cadmus in a year across all Cadmus modules, including students who have used Cadmus in multiple modules (i.e. more than once) is estimated by multiplying the average number of students enrolled in a Cadmus subject by the number of Cadmus modules. **Course withdrawal reduction rate** • Percentage-point improvement in pre-census course withdrawal rate between Cadmus and non-Cadmus cohorts. **Attribution factor** • Two-thirds of the observed withdrawal reduction is assumed as attributable to Cadmus. **Proportion of course withdrawals that do not result in a programme attrition** • Three-quarters of pre-census course withdrawals are assumed to not result in a degree/programme attrition.
Assumptions and limitations	**Assumptions** • Weighted average tuition fee is assumed to be uniform across institution, modules, disciplines, and student cohorts for the purposes of estimation. • No account for module re-enrolment, replacement enrolments (as the semester has already commenced), or deferred completion; assumes permanent loss of tuition revenue. **Limitations** • Weighted fee averages mask variation across disciplines and student groups; actual benefit may differ significantly by subject.	**Assumptions** • Modelling applies the observed average withdrawal reduction rate (2.7%) consistently across all modules and year levels and for all institutions. • Cadmus is assumed to be responsible for the majority (two-thirds) of withdrawal reduction given the pre- and post-Cadmus outcomes data collected by universities. Cadmus is not responsible for 100% of withdrawal reduction – other institutional factors (academic support, peer engagement, cohort characteristics) will have contributed. This proportion was selected to recognise the consistency of Cadmus-linked withdrawal reductions observed across multiple pilots, while acknowledging that not all observed improvements can be attributed solely to the platform. • The majority (three-quarters) of pre-census course withdrawals do not result in a full degree attrition – this reflects students' study load management or subject preference changes rather than attrition, and therefore only a temporary loss in revenue. It is assumed that a quarter of course withdrawals relate to a programme attrition, and a permanent loss in revenue for the university – this saving is excluded here to avoid double counting between the course withdrawal reduction and increases in retention (benefit #2). **Limitations** • The actual level of withdrawal reduction attributable to Cadmus cannot be determined with certainty by the university studies without any further detailed student-level analysis. The factor applied is a Nous estimate.
Data sources	• Australia: Department of Education indexed tuition rates (2026); Higher Education Statistics 2024; Nous analysis of domestic/international fee mix. • UK: Office for Students maximum fees (2025–26); Nous analysis of international fees.	• Number of students per university (as an input for number of Cadmus module enrolments): Australia Higher Education Statistics (2024), UK HESA Student and Staff Data (2024). • Course withdrawal reduction rates: Derived from pilot evaluation reports across three universities. • Attribution factor: No peer-reviewed data source available from education sector literature to directly estimate the causal attribution of platform improvements to withdrawal reduction.

Benefit 2: Lifetime value of retained students

We developed a calculation logic to quantify the lifetime value of retained students.

Student retention delivers significant long-term value for universities, strengthens learning outcomes and enhances the overall student experience. It enables universities to provide consistent teaching, maintain cohort continuity, and reduce the administrative load associated with student withdrawal. Continued enrolment also brings associated tuition revenue that accumulate over the latter stages of retained students' study.

Retention is strongly correlated with academic performance. Research shows that improvements in early academic outcomes – such as first-year grades – substantially increase the likelihood of completing a degree. For example, international studies indicate that each incremental improvement in GPA can translate into a markedly higher probability of credential attainment[1]. While these findings originate from U.S. contexts, the principle applies broadly: better grades early in a programme reduce attrition risk and strengthen progression.

Cadmus cannot easily collect student-level retention data, which is a limitation of this analysis. Because Cadmus is implemented at the subject level for all enrolled students – not taken up voluntarily by individuals – it is difficult to isolate the participation effects over the life of a degree or establish a natural comparison group. As a result, this analysis estimates retention uplift indirectly, translating observed grade improvements into a retention uplift rate.

Cadmus is designed to support these outcomes by improving assessment clarity, scaffolding tasks, and fostering engagement, which collectively drive grade uplift. Higher grades in first-year units signal stronger academic confidence and capability, reducing the risk of degree attrition and increasing persistence through to completion. This ROI modelling uses this relationship to approximate retention uplift based on observed improvements in average grades, translating academic success into measurable lifetime value for universities.

The logic model for calculating the lifetime value of retained students is presented below.

NET SURPLUS OF RETAINED STUDENT		INCREASE IN RETAINED STUDENTS		
Marginal benefit of retained student – Marginal cost of retained student	X	Number of unique students who have used Cadmus **x** Retention uplift **x** Attribution factor	=	LIFETIME VALUE OF RETAINED STUDENTS

The calculation details, assumptions, limitations and data sources for each of the components of our calculation logic are explained on the following page.

1. Grade-to-retention relationship: Turk (2024), Identifying Predictors of Credential Completion Among Beginning Community College Students; American Council on Education.

Approach to calculating lifetime value of retained students, assumptions and limitations.

	Net surplus of retained student	Increase in retained students
Calculation detail	**Marginal benefit of retained student** • Future tuition revenue per retained student is calculated by undergraduate year group (1st, 2nd, 3rd year) as the weighted average annual tuition fee multiplied by remaining years of study (assumes a 3-year degree). Weighted average tuition fee is calculated using the same inputs and assumptions as in the benefit per prevented withdrawal calculation (see page 10). • Remaining years are 2.5 (1st year), 1.5 (2nd year), and 0.5 (3rd year), which converts annual tuition into "lifetime" tuition remaining for that student. **Marginal cost of retained student** • Marginal cost per retained student is calculated as the marginal cost to the university per student per year ($14,249 in Australia, £6,596 in the UK, an average of multiple figures in the literature) multiplied by remaining years of study (by year group). • Marginal cost per student per year is derived from the average cost of "teaching and scholarship" per student, adjusted by a marginal cost factor to reflect that not all average costs are incurred with an additional student.	**Number of unique students who have used Cadmus** • Number of students identified by Cadmus platform. **Retention uplift** • Retention uplift is calculated by translating observed Cadmus-linked grade improvements (%) into an estimated degree-level retention uplift (%). • Retention uplift is estimated by year group (higher retention uplift in first year, lower in 2nd and 3rd years). **Attribution factor** • 25% of the observed withdrawal reduction is attributed to Cadmus, applied to all year groups.
Assumptions and limitations	**Assumptions** • Retained students pay fees for all remaining study periods. • Average cost across all students reflects the average cost across Studiosity users (i.e. no variation by domestic/internation al, faculty etc.). • The marginal cost factor approximates that the majority (67%) of the average cost of teaching and scholarship is variable (e.g., incurred with an additional student, such as marking) and that some costs are fixed (e.g., facilities). **Limitations** • Marginal revenue and cost of a retained student does not segment by discipline, delivery mode or student type, which can materially shift revenue and cost. • There is no accepted literature on a typical marginal cost for an additional student as the costs are highly variable and non linear (e.g. one additional student being placed in an existing class vs. a new class being added to accommodate additional students). • While average cost research is reported in the literature, an average cost is not reflective of the 'marginal cost of teaching' incurred from an additional student and is likely substantially higher than the marginal cost of an additional student. This is the justification for using a marginal cost factor.	**Assumptions** • Retention uplift is estimated based on academic evidence of relationships between GPA and credential completion. Turk (2024) identified that a 1.0 increase in GPA improves retention by 27%. 1.0 GPA is estimated to be equivalent to 15% in % mark terms. Retention uplift is calculated to be 1.4% per 1% average grade improvement, including a 25 per cent discount factor as the Turk figure is based on correlation and not causal evidence. • Retention uplift benefit from Cadmus relative to 1st year is assumed by Nous to be 25% in 2nd year and 10% in 3rd year as attrition risk falls. • Cadmus is assumed to be responsible for a proportion (25%) of retention uplift – other factors (academic support, peer engagement, cohort quality) will have contributed significantly. The 25% attribution factor for retention uplift is intentionally conservative and accounts for the fact that the underlying grade-to-retention relationship is based on correlational research rather than direct causal evidence. **Limitations** • The actual level of retention uplift attributable to Cadmus cannot be determined with certainty by the university studies without any further detailed student-level analysis. The factor applied is a Nous estimate.
Data sources	• Tuition fees: Department of Education indexed tuition rates (2026); UK: Office for Students maximum fees (2025–26). • Average cost: averages calculated from Times Higher Education 2025 Australia university analysis, with proportion attributable to teaching and scholarship, 2026 adjusted Australian Government Department of Education Transparency in Higher Education Expenditure report (2019), and 5Rs Partnership 2025 Fee Monitor data	• Number of unique students per Cadmus module: Collected from Cadmus pilot studies. • Average grade improvements (as an input for retention uplift): Derived from pilot evaluation reports across 11 universities. • Grade-to-retention relationship: Turk (2024), Identifying Predictors of Credential Completion Among Beginning Community College Students; American Council on Education.

Estimated ROI

Modelling estimates a positive ROI for universities using Cadmus – a weighted average of 7.1 among pilots, growing significantly with deeper deployment.

Cadmus' ROI represents the net result of the estimated annual financial benefit to a university compared to the annual subscription fee of Cadmus. Modelling quantifies two benefit streams for universities: reduced pre-census course withdrawals (which reduce tuition revenue), and retention uplift (where improved academic performance reduces attrition rates/lifts retention rates).

The weighted average ROI among pilots at Australian and UK universities is 7.1.

Results across pilots at Australian and UK universities are presented in the table below. The range of ROIs reflects the variation in number of Cadmus modules and in the number of students using Cadmus. Cadmus' estimated ROI increases as universities deploy Cadmus across more and larger modules, with benefits scaling with the number of students reached. Given that many universities have deployed more modules post-pilot, the ROI at those universities is expected to have already increased. When analysing each quantified benefit in isolation, the average ROI also remains positive. Retention benefits comprise approximately 60% of total estimated benefits to each university.

ROI estimates exclude other non-quantified benefits, such as time savings and quality improvements for academics designing, marking, and managing assessments.

	ROI
Range for Australian pilots (5 – 108 modules)	1.2 – 19.6
Average Australian pilot	**8.6**
Range for UK pilots (10 – 63 modules)	-0.3 – 5.7
Average UK pilot	**4.3**
Weighted average pilot	**7.1**

Standardised estimates for ROIs for a prospective Australian and UK university using Cadmus for 50 modules are presented below. The standardised Australian ROI is higher than the UK ROI, because the Australian archetype assumes a larger student base, a higher student-to-teacher ratio, and a larger net margin per retained student.

	Standardised AUS	Standardised UK
Pilot of 50 modules	**5.9**	**2.7**

Key assumptions:

- **Standardised university characteristics:** Modelling standardises Australian and UK institutional profiles (student demographics, tuition fees, teaching loads, withdrawal, retention and degree-completion dynamics).
- **Undergraduate focus:** Modelling is based on modules offered to undergraduates, given this is the focus and there is no further data on different benefits for postgraduate students. Retention uplift value uses a 3-year undergraduate degree, with the uplift in retention declining in each year of study as retention risk falls.
- **Inclusion of average student outcomes:** ROI modelling assumes Cadmus delivers consistent withdrawal rate reductions (2.7% pre-census withdrawal reduction rate) and academic improvement (2.7% average grade improvement) across pilot institutions. Actual outcomes vary by discipline, cohort, and university.
- **Attribution factors discount benefits:** Benefits are discounted by attribution factors (67% for withdrawal reduction, 25% for retention uplift) to acknowledge that not all measured outcome improvement is attributable to Cadmus alone. This is a conservative methodological choice.

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PEOPLE	PRINCIPALS	OFFICES